A permutation-character theorem for quadratic wild-place discriminant ratios

A research note toward Part IV of the metaplectic Jacquet–Langlands series

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Abstract

For a quadratic extension E/F of nonarchimedean local fields with odd residue characteristic, and any odd integer n for which $\mu_n \subset F$, we prove that the ratio of the Weyl discriminants $\Delta(t)/\Delta(t^n)$, viewed as a function on $E^\times/F^\times$, is a finite permutation character. This settles the *toral positivity* step proposed in Part IV, Draft 5, for $r = 2$ at every wild place. It does not prove irreducible-character integrality for the full covering group $\widetilde{D^\times}$; the nonabelian gluing problem remains open.

1 The theorem

Let F be a finite extension of $\mathbb{Q}_p$ with p odd, residue field of cardinality q, and let $n \geq 1$ be odd with $\mu_n \subset F$. Let E/F be any quadratic field extension, with nontrivial Galois involution σ. Put

$$A = E^\times/F^\times, \qquad G = E^1 = \ker(N_{E/F} : E^\times \to F^\times).$$

Hilbert's Theorem 90 identifies $A \xrightarrow{\sim} G$ via $t \mapsto u = t/\sigma(t)$. The normalized absolute value on E is $|x|_E = q_E^{-v_E(x)}$, so $|x|_E = |N_{E/F}x|_F$.

For noncentral $t \in E^\times$ let

$$R_E(t) = \frac{\Delta(t)}{\Delta(t^n)}, \qquad \Delta(t) = \left|\det(1 - \mathrm{Ad}(t) \mid D/E)\right|_F^{1/2}.$$

The ratio extends continuously at scalar elements, with value $|n|_F^{-1}$.

Theorem 1 (Quadratic toral permutation positivity)**.** *For any $E/F, n$ as above, R_E is the character of a finite-dimensional permutation representation of a finite quotient of A. In particular,*

$$R_E(t) = \sum_{\chi \in \widehat{A}} m_\chi \chi(t), \qquad m_\chi \in \mathbb{Z}_{\geq 0},$$

where only finitely many m_χ are nonzero. This includes every wild case $p \mid n$. At the identity $R_E(1) = |n|_F^{-1}$.

2 Valuation formula

Write $n = p^a m$ with $(m, p) = 1$, and put

$$t_\zeta = v_E(1 - \zeta), \qquad G_k = \{u \in G : v_E(1 - u) \geq k\} \quad (k \geq 1), \qquad G_0 = G.$$

If $a = 0$ then $R_E \equiv 1$, so assume $a \geq 1$.

Lemma 2. *For $u \in G$ and any $\zeta \in \mu_n$ whose prime-to-p component is nontrivial, $|1-\zeta u|_E = 1$. For $\zeta \in \mu_{p^a} \setminus \{1\}$,*

$$v_E(1-\zeta u) = \min\{v_E(1-u), t_\zeta\}.$$

Consequently, with $k = v_E(1-u) \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$,

$$R_E(u) = q_E^{\frac{1}{2}\sum_{\zeta \in \mu_{p^a}\setminus\{1\}} \min(k, t_\zeta)}. \tag{1}$$

Proof. The reduction of $u \in E^1$ lies in a norm-one subgroup. In the unramified case its intersection with $k_F^\times$ is $\{\pm 1\}$, and in the ramified case the reduction of u is itself in $\{\pm 1\}$. A root of unity with nontrivial prime-to-p component has reduction in $k_F^\times \setminus \{\pm 1\}$, since n is odd, proving the first assertion.

For a nontrivial p-power root ζ, the ultrametric inequality gives the asserted minimum when the two valuations differ. If both equal t_ζ and $v_E(1-\zeta u) > t_\zeta$, then $u \equiv \zeta^{-1} \pmod{\mathfrak{p}_E^{t_\zeta+1}}$. Applying σ and using $\sigma(u) = u^{-1}$ gives $u^{-1} \equiv \zeta^{-1}$ to the same precision, hence $\zeta \equiv \zeta^{-1}$, a contradiction to $v_E(\zeta - \zeta^{-1}) = t_\zeta$ (as p is odd). Thus equality holds.

For $u = t/\sigma(t)$ we have $\Delta(t) = |N_{E/F}(1-u)|_F^{1/2} = |1-u|_E^{1/2}$. Factor $1 - u^n = \prod_{\zeta \in \mu_n}(1-\zeta u)$ and use the two valuation statements. The identity follows for noncentral t and extends continuously to $u = 1$. At $u = 1$, the cyclotomic product $\prod_{\zeta \in \mu_{p^a}\setminus\{1\}}(1-\zeta) = p^a$ gives $R_E(1) = |p^a|_E^{-1/2} = |n|_F^{-1}$. □

Define for $k \geq 0$

$$M_k = q_E^{\frac{1}{2}\sum_{\zeta \in \mu_{p^a}\setminus\{1\}} \min(k, t_\zeta)}, \qquad M_0 = 1,$$

and put $T = \max_{\zeta \neq 1} t_\zeta$. All values M_k are positive integers because roots ζ, ζ^{-1} occur in pairs of equal valuation. We have $M_k = M_T$ for $k \geq T$.

3 Unramified quadratic extensions

Suppose E/F is unramified. Then $q_E = q^2$ and reduction of the norm-one group has order $q+1$; at every deeper level the trace-zero subgroup of the residue-field additive group has order q. Thus the filtration satisfies

$$[G : G_1] = q+1, \qquad [G : G_k] = (q+1)q^{k-1} \quad (k \geq 1).$$

Write $s_k = \#\{\zeta \in \mu_{p^a} \setminus \{1\} : t_\zeta \geq k\}$. Each s_k is even, and $M_k/M_{k-1} = q^{s_k}$. Set

$$c_k = \frac{M_k - M_{k-1}}{(q+1)q^{k-1}} \quad (1 \leq k \leq T).$$

For $k = 1$, $M_0 = 1$ and $q+1$ divides $q^{s_1} - 1$ because s_1 is even. For $k > 1$ with $s_k > 0$, M_{k-1} is divisible by q^{k-1}, since at least two roots contribute at least $k-1$ to its exponent. Again $q+1$ divides $q^{s_k} - 1$. Hence $c_k \in \mathbb{Z}_{\geq 0}$ (and $c_k = 0$ if $s_k = 0$).

By telescoping (1),

$$\boxed{R_E = \mathbf{1}_G + \sum_{k=1}^{T} c_k \, [G : G_k] \, \mathbf{1}_{G_k}.} \tag{2}$$

Here $[G : G_k]\,\mathbf{1}_{G_k}$ is precisely the permutation character of G on G/G_k.

4 Ramified quadratic extensions

Suppose E/F is ramified. Then $q_E = q$ and $t_\zeta = 2v_F(1-\zeta)$ is even. The norm-one group has filtration

$$[G : G_1] = 2, \qquad G_{2j} = G_{2j+1}, \qquad [G : G_{2j}] = 2q^j \quad (j \geq 1).$$

For completeness, write $E = F(\sqrt{d})$ with $v_F(d) = 1$. Hilbert 90 gives $u = (a+b\sqrt{d})/(a-b\sqrt{d})$; normalize a, b to be integral and not both divisible by the uniformizer of F. If a is a unit, scale it to 1 and write $c = b/a \in \mathcal{O}_F$: then the depth of $u-1$ is $1 + 2v_F(c)$. Moreover, two such parameters c, c' define elements congruent modulo $\mathfrak{p}_E^{2j}$ precisely when $c \equiv c' \pmod{\mathfrak{p}_F^j}$, so $[G_1 : G_{2j}] = q^j$. If a is not a unit and b is, then $u \equiv -1$ modulo $\mathfrak{p}_E$. These observations give the displayed indices and the fact that positive depths are odd.

Define

$$c_1 = \frac{M_1 - 1}{2}, \qquad c_{2j} = \frac{M_{2j+1} - M_{2j-1}}{2q^j} \quad (1 \leq j \leq T/2).$$

The first coefficient is an integer because $M_1 = q^{(p^a-1)/2}$ is odd. For $j \geq 1$ write

$$\frac{M_{2j+1}}{M_{2j-1}} = q^{e_j}, \qquad e_j = \frac{1}{2}\sum_{\zeta \neq 1} (\min(2j+1, t_\zeta) - \min(2j-1, t_\zeta)) \in \mathbb{Z}_{\geq 0}.$$

If $e_j = 0$, then $c_{2j} = 0$. Otherwise some pair $\{\zeta, \zeta^{-1}\}$ has $t_\zeta \geq 2j$, so M_{2j-1} is divisible by q^{2j-1}, hence by q^j. Moreover $q^{e_j} - 1$ is even. Therefore $c_{2j} \in \mathbb{Z}_{\geq 0}$.

As all positive depths are odd, telescoping now gives

$$\boxed{R_E = \mathbf{1}_G + c_1\,[G : G_1]\mathbf{1}_{G_1} + \sum_{j=1}^{T/2} c_{2j}\,[G : G_{2j}]\,\mathbf{1}_{G_{2j}}.} \tag{3}$$

Again each summand is the character of a permutation representation. Equations (2) and (3) prove Theorem 1.

5 Examples and scope

For $F = \mathbb{Q}_3(\zeta_3)$, $n = 3$, the unramified case has $q = 3, T = 1, M_1 = 9, c_1 = 2$ and

$$R_E = 1 + 2\,\mathrm{Reg}(C_4).$$

For either ramified quadratic extension, $q = 3, T = 2, M_1 = 3, M_3 = 9, c_1 = c_2 = 1$, giving

$$R_E = 1 + \mathrm{Reg}(C_2) + \mathrm{Reg}(C_6),$$

recovering Part IV, Draft 5. As a further illustration, for $F = \mathbb{Q}_5(\zeta_5)$ and $n = 5$, the same proof gives $1 + 104\,\mathrm{Reg}(C_6)$ in the unramified case, and $1 + 12\,\mathrm{Reg}(C_2) + 60\,\mathrm{Reg}(C_{10})$ in the ramified case.

Remark 3 (What is not proved)**.** *By the toral Fourier criterion of Part IV, Draft 5, the theorem supplies actual character restrictions for transfers of any ordinary irreducible representation to each maximal covered field torus when $r = 2$. However, positivity on every torus need not glue to a positive-definite function on the full nonabelian group. An additional argument, such as nonabelian finite-level positive-semidefiniteness, a Brauer-type virtual-character criterion, or an explicit induced-representation construction, is essential for wild-place irreducible-character integrality.*

Remark 4 (Positivity and the unitary character matrix)**.** *The referee's observation is useful: if L_D is unitary on the irreducible-character Hilbert bases and* every *transferred irreducible character is positive definite, then the matrix of L_D has entrywise nonnegative real coefficients. Distinct rows are orthogonal with disjoint supports; column norm one forces a single entry equal to one in every column. Thus the full matrix is a permutation matrix. Positivity of only one row does not suffice: a unit vector with several positive entries need not be integral.*